📌 PHASE 1 — Backend Project Setup

Objective

Setup backend architecture and environment.

🛠️ Technologies

- Python
- Flask
- MySQL
- SQLAlchemy (optional ORM)
- OpenPyXL
- Pandas
- APScheduler

✅ Tasks

Install Backend Libraries

Flask
flask-cors
mysql-connector-python
openpyxl
pandas
APScheduler

Create Backend Folder Structure

```
backend/
 |
 ├── app.py
 ├── config.py
 ├── routes/
 ├── database/
```

├── models/
├── scheduler/
├── services/
├── exports/

Configure Flask Server

Features:

- API setup
- CORS setup
- JSON responses
- error handling

Setup Environment Variables

Store:

- DB credentials
- secret keys
- email configs

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- Flask server running
- MySQL connection established
- Base project structure ready

📌 PHASE 2 — Database Design & Setup

Objective

Create and manage project database.

✅ Tasks

Create Database

Database Name

attendance_system

Create Tables

Students Table

Column	Type
id	INT
name	VARCHAR
roll_no	VARCHAR
department	VARCHAR
semester	VARCHAR
email	VARCHAR
face_encoding	TEXT
created_at	DATETIME

Attendance Table

Column	Type
id	INT
student_id	INT
date	DATE
time	TIME

Column	Type
status	VARCHAR

Admin Table

Column	Type
id	INT
username	VARCHAR
password	VARCHAR

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- Database schema completed
- Tables created
- Relationships working

📌 PHASE 3 — Authentication System

Objective

Create secure admin authentication.

✅ Tasks

Admin Login API

Features:

- username/password validation
- session/token creation

Authentication Middleware

Protect:

- dashboard routes
- export routes
- student management routes

Logout System

Destroy session/token.

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- Working authentication system

📌 PHASE 4 — Student Management APIs

Objective

Handle student registration and management.

✅ Tasks

Create APIs

Register Student API

Receives:

- student details
- face encoding

Stores data in database.

Fetch Students API

Returns:

- all students

- filters
- pagination

Update Student API

Allows admin editing.

Delete Student API

Removes:

- student
- attendance records

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- Complete CRUD APIs

📌 PHASE 5 — Attendance Management System

Objective

Handle attendance storage and validation.

✅ Tasks

Attendance API

Receives:

- recognized student ID
- timestamp

Stores attendance.

Duplicate Attendance Prevention

Logic:

```
if already_marked_today:
    reject_attendance()
```

Very important feature.

Attendance Filtering APIs

Filter by:

- date
- department
- student

Attendance Status APIs

Returns:

- present students
- absent students
- total attendance

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- Attendance system working
- Duplicate prevention implemented

📌 PHASE 6 — Excel Automation System

Objective

Automatically generate and update attendance Excel sheets.

✅ Tasks

Create Monthly Excel Files

Example:

attendance_may_2026.xlsx

Update Excel Automatically

When attendance marked:

Backend:

- opens Excel file
- updates correct student row
- writes:

P (09:14 AM)

Auto Create Date Columns

If new day:

- create new column automatically

Generate Reports

Export:

- monthly attendance
- filtered attendance
- department reports

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- Excel automation fully working

📌 PHASE 7 — Auto-Absent Scheduler

Objective

Automatically mark absent students after college hours.

✅ Tasks

Setup Scheduler

Using:

- APScheduler

Daily Automation Logic

At:

5:00 PM

System:

- checks students without attendance
- marks them absent
- updates Excel

Prevent Repeated Execution

Scheduler should:

- run once daily only

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- Auto-absent system operational

📌 PHASE 8 — Email Report System

Objective

Automatically send attendance reports.

✅ Tasks

Email Functionality

Send:

- daily report
- monthly report

To:

- admin/faculty email

Attach Excel File

System sends:

attendance_may_2026.xlsx

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- Email reporting system working

📌 PHASE 9 — Dashboard APIs

Objective

Provide analytics data to frontend dashboard.

✅ Tasks

Analytics APIs

Return:

- total students
- present count
- absent count
- attendance percentage

Graph APIs

Provide:

- weekly data
- monthly trends
- department analytics

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- Dashboard APIs completed

📌 PHASE 10 — Backend Optimization & Testing

Objective

Stabilize backend before final demo.

✅ Tasks

Error Handling

Handle:

- invalid requests
- DB errors
- duplicate entries

API Testing

Test all APIs using:

- Postman

Performance Testing

Ensure:

- fast attendance marking
- stable Excel updates

Logging System

Log:

- attendance activity
- errors
- unknown events

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- Stable production-ready backend